Worksheet 43 · Similar Shapes IGCSE

Edexcel IGCSE Mathematics · past paper questions

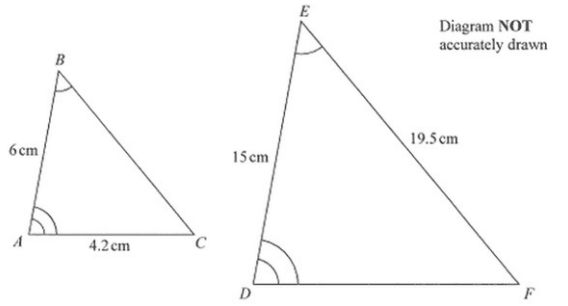
Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Identify similar shapes • Use scale factors to find missing lengths • Solve problems involving similar triangles		
Grade: / 34	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Similar Shapes*

Question 1 (4 marks)

ABC and *DEF* are similar triangles.



(a) Work out the length of *DF*.

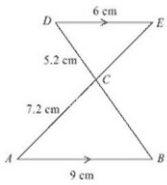
..... cm (2)

(b) Work out the length of *BC*.

..... cm (2)

Question 2 (4 marks)

AB is parallel to *DE*. *ACE* and *BCD* are straight lines. *AB* = 9 cm. *AC* = 7.2 cm. *CD* = 5.2 cm. *DE* = 6 cm.



(a) Calculate the length of *BC*.

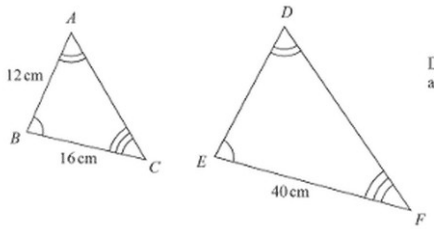
..... cm (2)

(b) Calculate the length of *CE*.

..... cm (2)

Question 3 (4 marks)

ABC and *DEF* are similar triangles.



(a) Work out the length of *DE*.

..... cm (2)

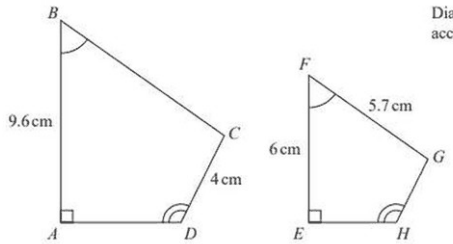
The area of triangle *DEF* is 525 cm²

(b) Find the area of triangle *DEF* in m²

..... m² (2)

Question 4 (4 marks)

ABCD and *EFGH* are similar quadrilaterals.



(a) Work out the length of *GH*

..... cm (2)

(b) Work out the length of *BC*

..... cm (2)

Question 5 (4 marks)

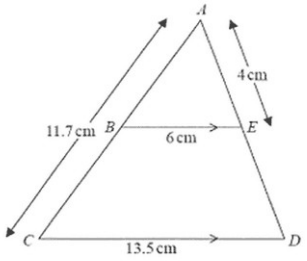


Diagram **NOT** accurately drawn

The diagram shows triangle ACD.
B is a point on AC and E is a point on AD so that BE is parallel to CD.

$AE = 4$ cm
$AC = 11.7$ cm
$BE = 6$ cm
$CD = 13.5$ cm

(a) Calculate the length of AB.

..... cm
(2)

(b) Calculate the length of ED.

..... cm
(2)

Question 6 (2 marks)

ABC and DEF are similar triangles.

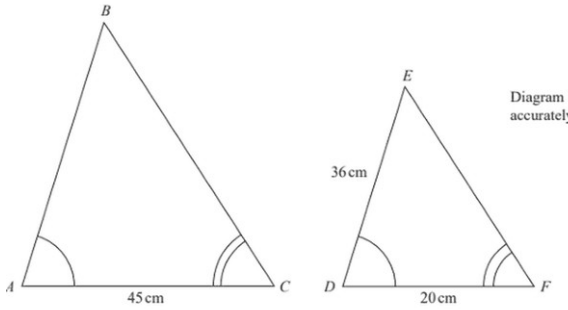


Diagram **NOT** accurately drawn

(a) Work out the length of AB.

..... cm
(2)

Given that $BC = 54$ cm,

(b) work out the length of EF.

..... cm
(2)

Question 7 (3 marks)

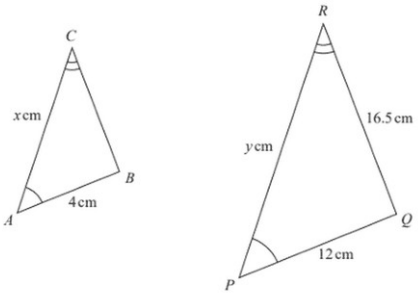


Diagram **NOT** accurately drawn

Triangle ABC is similar to triangle PQR

$AB = 4$ cm $PQ = 12$ cm $RQ = 16.5$ cm $AC = x$ cm $PR = y$ cm

(a) Calculate the length of BC

..... cm
(2)

(b) Write down an expression for y in terms of x

$y =$ .....
(1)

Question 8 (2 marks)

The diagram shows two water towers in Kuwait.

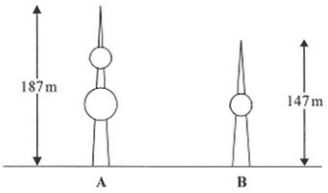


Diagram **NOT** accurately drawn

The real height of tower **A** is 187 m.
The real height of tower **B** is 147 m.

Ahmed makes a scale model of both towers.

The height of tower **A** on the scale model is 90 cm.

Work out the height of tower **B** on the scale model.
Give your answer correct to the nearest centimetre.

..... cm

Question 9 (3 marks)

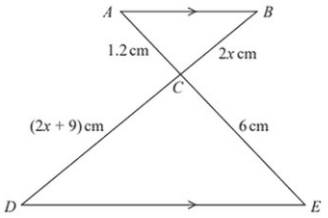


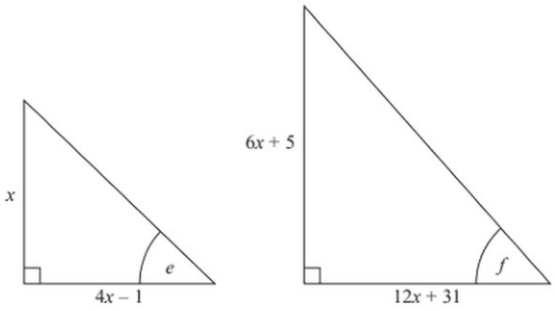
Diagram **NOT** accurately drawn

ACE and *BCD* are straight lines.
AB is parallel to *DE*

Work out the value of x

Question 10 (4 marks)

Here are two right-angled triangles.



Given that

$$\tan e = \tan f$$

find the value of x.

You must show all your working.